Manufacturing Decision Copilot

Hackathon Submission Report

Track	Manufacturing Procurement Intelligence
Primary URL	https://manufacturing-decision-copilot-farhan.vercel.app/dashboard
Version	1.0 · August 2026
Backend	FastAPI 0.141 · Python 3.12 · PostgreSQL 16 · ChromaDB 0.5 · Celery 5 · Redis 7
Frontend	Next.js 15 App Router · TypeScript · Tailwind CSS v4 · shadcn/ui
AI Stack	PydanticAI · Gemini 2.0 Flash · BAAI/bge-small-en-v1.5 (local embeddings)
Unit Tests	501 passing · 0 failures · 2.50 s runtime
Suppliers	10 across 7 countries · 17 challenge pack documents · 3 scenario simulations

Scan to open the live application
https://manufacturing-decision-copilot-farhan.vercel.app/dashboard

The dashboard loads with all 10 suppliers pre-seeded. Recommendation, scenario engine, and evidence citations are all live.

Design principle: AI explains. Math decides. Humans approve.

1. What We Built

Manufacturing Decision Copilot transforms fragmented procurement documents into a clear, evidence-backed supplier recommendation — in under 30 seconds. It is not a chatbot. It is a decision intelligence platform built around one core principle: **the LLM never touches a number**. All scoring is pure, deterministic Python. The AI reads sanitised document evidence and explains what the math found.

Core Capabilities

Capability	Description
Document Ingestion	PDF/DOCX/XLSX upload → PyMuPDF extraction → semantic chunking → ChromaDB embedding
6-Dimension Scoring	cost · quality · delivery · risk · capability · compliance — all pure Python
Hybrid RAG	ChromaDB cosine similarity + BM25Okapi + Reciprocal Rank Fusion → top-K evidence
Self-RAG Loop	Up to 3 retrieval iterations; LLM grader refines query if coverage < 0.025
Scenario Engine	Shipping shock · tariff escalation · supplier removal · cert override — <2 ms
Confidence Engine	5-factor deterministic formula (0–100%); LLM never produces this number
AI Explanation	5 typed PydanticAI agents; Pydantic v2 schema enforcement; hallucination filtering
Executive Report	AI-generated board-ready PDF with citations, risk summary, next steps
Evidence Citations	Every AI statement linked to exact PDF page + chunk; clickable in the dashboard

2. Architecture and Data Flow

The system is built on a strict layered architecture. Routes validate and delegate. Services orchestrate. Repositories handle I/O. Engines perform pure math. AI is called only through services — never directly from routes.

```
Browser / Next.js 15 (Vercel)
■ REST /api/v1
▼
FastAPI Backend (Railway)
■■■ API Routes validate + call service
■■■ Services business workflow
■■■ Repositories DB I/O only
■■■ Engines pure Python math ← NO AI, NO I/O
■ cost · quality · delivery · risk
■ compliance · capability · ranking
■ scenario · confidence
■■■ AI Layer PydanticAI → Gemini / GPT-4o
■ retriever (hybrid BM25+vector+RRF+Self-RAG)
■ guardrails (7 injection patterns stripped)
■ 5 typed output schemas (Pydantic v2 validated)
■■■ Workers Celery + Redis
PDF → chunks → embeddings → ChromaDB
→ ExtractionAgent → PostgreSQL

Databases
PostgreSQL 16 source of truth (10 tables, Alembic migrations)
ChromaDB 0.5 vector index (document_chunks collection)
Redis 7 job queue · recommendation cache (TTL 5 min)
```

Hero Data Flow

```
1 Upload PDF → MIME validate · SHA-256 checksum · /uploads/{uuid}.pdf
2 Celery worker → PyMuPDF extract → 800-token semantic chunks
3 Embed chunks → BAAI/bge-small-en-v1.5 (local, 384-dim)
4 Store in ChromaDB → metadata: project_id · supplier_id · page · section
5 ExtractionAgent → SupplierExtraction schema → PostgreSQL fields
6 Get Recommendation → score_suppliers() [pure Python, ~2 ms]
7 retrieve_with_loop() → ChromaDB cosine (top 20) + BM25 (top 20)
→ RRF fusion → top 8 chunks
→ LLM grader grades coverage → refine if needed
8 RecommendationAgent → sanitised chunks + scores → narrative + evidence_ids
9 filter_evidence_ids()→ strip any hallucinated chunk references
10 calculate_confidence → deterministic 5-factor formula (NEVER the LLM)
11 Store in PostgreSQL → recommendations + recommendation_evidence + audit_log
12 Cache in Redis → TTL 5 min
13 Dashboard renders → DonutGauge · CitationPopover · ScenarioSimulator
```

3. Scoring Engines — Exact Formulas

All 9 engines are pure stateless Python functions. Zero I/O. Zero AI. 501 unit tests, all passing. The LLM never reads, modifies, or influences any score.

Cost Engine

```
landed_cost = (quoted_price × material_mult
+ shipping × shipping_mult
+ duty + insurance + taxes) × currency_rate
cost_score = (min_landed_cost / supplier_landed_cost) × 100
```

Quality Engine

```
quality_score = defect_component × 0.40
+ inspection_pass_rate × 0.35
+ customer_rating_norm × 0.25
```

Delivery Engine

```
delivery_score = lead_time_component × 0.45
+ on_time_delivery_pct × 0.35
+ production_capacity_pct × 0.20
lead_time_component = (min_lead_time / supplier_lead_time) × 100
```

Risk Engine

```
risk_score = 100 − (financial_risk × 0.25
+ country_risk × 0.20
+ supply_risk × 0.20
+ compliance_risk × 0.20
+ capacity_risk × 0.15)
[inputs are risk magnitudes; DB stores safety scores → mapper inverts]
```

Compliance Engine

```
if any required_cert NOT IN supplier_certs:
compliance_score = 0.0 ← auto-disqualification
else:
compliance_score = 100.0
[prefix matching: 'ISO 9001' matches 'ISO 9001:2015']
```

Capability Engine

```
capability_score = match_score × 0.70
+ capacity_pct × 0.30
+ (10 if eng_support else 0)
match_score = (req_capabilities_met / total_required) × 100
```

Ranking Engine

```
final_score = cost_score × 0.30
+ quality_score × 0.20
+ delivery_score × 0.15
+ risk_score × 0.15
+ capability_score × 0.10
+ compliance_score × 0.10 [all weights configurable]
```

Confidence Engine

```
confidence = extraction_quality × 0.30
+ evidence_coverage × 0.20
+ retrieval_quality × 0.20
+ rule_agreement × 0.20
+ data_completeness × 0.10
→ 0.0–1.0 displayed as % with Low / Medium / High label
```

Scenario Engine

```
ScenarioConfig: shipping_multiplier · currency_rate
demand_multiplier · lead_time_adjustment_days
supplier_availability · certification_overrides
material_cost_multiplier · import_duty_rate
Run baseline + scenario through score_suppliers() → diff → AI explains
```

4. AI Layer and RAG Pipeline

4.1 Five PydanticAI Agents

Agent	Output Schema	Purpose
RecommendationAgent	RecommendationOutput	Explains top supplier: pros, cons, tradeoffs, risks, next actions, evidence_ids
ScenarioExplainerAgent	ScenarioExplanation	Plain-English explanation of what changed after a scenario simulation
ExecutiveSummaryAgent	ExecutiveSummary	Board-ready summary with recommendation statement, risk summary, next steps
ComparisonAgent	ComparisonOutput	Head-to-head two-supplier comparison with strengths/weaknesses
ExtractionAgent	SupplierExtraction	Structured field extraction from uploaded PDF chunks

4.2 Hybrid Retrieval Pipeline

```
Query string
■
■■■ BAAI/bge-small-en-v1.5 (local embeddings, 384-dim)
■ ▼
■■■ ChromaDB cosine similarity → top 20 chunks
■ (filtered: project_id + optional supplier_id)
■
■■■ BM25Okapi keyword search on same candidates → top 20
■
■■■ Reciprocal Rank Fusion score = Σ 1/(60 + rank)
■ ▼ deduplicated, sorted descending
■■■ top-K RankedChunks (default K=8)

Self-RAG Loop (retrieve_with_loop):
MAX_LOOPS = 3
COVERAGE_THRESHOLD = 0.025 (avg RRF score)
MIN_CHUNKS_REQUIRED = 3
After each pass → LLM grader returns {sufficient, refined_query}
Stops when: grader says sufficient OR coverage ≥ threshold OR max loops
```

4.3 Guardrails

Pattern	Regex Target	Replacement
ignore_previous	ignore (all)? previous instructions	[REDACTED]
you_are_now	you are now a/an [not MDC/Copilot]	[REDACTED]
new_instructions	new/updated/revised instructions:	[REDACTED]:
system_prompt_leak	print/reveal/show your system prompt/rules	[REDACTED]
role_injection	<system> <user> <assistant> <ai> tags	[TAG]
act_as	act as a/an [word]	[REDACTED]
jailbreak_dan	DAN / jailbreak / unfiltered / unrestricted	[REDACTED]

All chunk text truncated to 4,000 chars before LLM call. LLM output validated against Pydantic schema — violation triggers Redis fallback. Evidence IDs hallucinated by LLM stripped before DB write.

5. Dataset and Data Manifest

5.1 10 Suppliers — Motor Housing Component

Supplier	Country	Landed Cost	Lead Time	Risk Level	Certifications	Edge Case
FastTrack Manufacturing	Mexico	$112.00	10d	Low	ISO+RoHS+IATF	Baseline #1
Acme Precision Mfg	Germany	$121.75	14d	Low	ISO+AS9100D+RoHS	Aerospace cert
NovaCast Engineering	Poland	$117.95	12d	Low	ISO+AS9100D	EU near-shore
VoltEdge Components	S. Korea	$132.48	16d	Low	ISO+IATF	Quality leader
TechForge Industries	Taiwan	$139.54	21d	Low	ISO+RoHS	Premium quality
Reliable Parts Co	India	$133.07	18d	Moderate	ISO+IATF	Mid-range
AlphaForge Ltd	Canada	$107.30	8d	Moderate	NONE	Zero certs → score=0
SteelPath Industries	Brazil	$121.28	35d	Elevated	ISO	MOQ=2000, BB- credit
Global Fabrication Ltd	China	$146.25	28d	High	ISO	25% duty, missing certs
PeakMetal Solutions	Vietnam	$108.10	30d	High	EXPIRED ISO	Expired cert → score=0

5.2 17 Challenge Pack Documents

#	Document	Type	Purpose
1–10	Supplier quotation PDFs (one per supplier)	Quotation	Price, lead time, MOQ, Incoterms, certs
11	Acme_ISO9001_AS9100D_Certificate_2026.pdf	Certificate	AS9100D + ISO 9001 verification
12	TechForge_RoHS_Certificate_2026.pdf	Certificate	RoHS compliance
13	FastTrack_RoHS_Certificate_2026.pdf	Certificate	RoHS compliance
14	VoltEdge_ISO9001_Certificate_2026.pdf	Certificate	ISO 9001 verification
15	MotorHousing_Technical_Specification_v3.pdf	Tech Spec	Required capabilities and tolerances
16	Purchase_Requirements_FY2027.pdf	Requirements	Procurement constraints, timeline
17	NovaCast_Supplier_Audit_Report_2026.pdf	Audit Report	Third-party risk evidence for NovaCast

5.3 3 Pre-Defined Scenarios

Scenario	Config Change	Expected Outcome
Baseline — Default Weights	No changes	FastTrack #1 at 93.37
Shipping Shock — Freight +40%	shipping_multiplier=1.4, lead_time +7d	FastTrack retains #1 (+0.15 delta)
China Tariff +50% Additional Duty	duty=0.75 for China; GlobalFab disabled	Acme Precision #1 (90.03); GlobalFab=0

6. Quantitative Evaluation Results

All scores below are produced by running the live deterministic engines against the seeded dataset. Every number is fully reproducible.

6.1 Baseline Ranking

Rank	Supplier	Final	Cost	Quality	Delivery	Risk	Cap.	Comp.	Landed $
1	FastTrack Manufacturing	93.37	95.8	94.1	86.2	85.9	100	100	$112.00
2	Acme Precision Mfg	91.89	88.1	97.5	78.7	94.4	100	100	$121.75
3	NovaCast Engineering	90.61	91.0	93.8	79.0	84.8	100	100	$117.95
4	VoltEdge Components	87.60	81.0	96.3	73.8	86.5	100	100	$132.48
5	TechForge Industries	84.74	76.9	95.3	67.0	83.7	100	100	$139.54
6	Reliable Parts Co	83.32	80.6	91.2	67.5	75.4	95	100	$133.07
7	AlphaForge Ltd	79.77	100.0	88.7	89.4	63.9	90	0	$107.30
8	SteelPath Industries	79.52	88.5	84.3	53.9	57.9	93	100	$121.28
9	Global Fabrication Ltd	77.02	73.4	85.9	57.6	65.3	94	100	$146.25
10	PeakMetal Solutions	70.02	99.3	81.5	49.9	51.5	87	0	$108.10

Green = baseline winner. Yellow = compliance=0 (zero certs). Red = compliance=0 (expired cert). AlphaForge has cheapest landed cost but zero certifications sink it to #7.

6.2 Scenario Comparison — Score Deltas

Supplier	Baseline	Shipping +40%	Δ Shock	China Tariff	Δ Tariff
FastTrack Manufacturing	93.37	93.52	+0.15	88.61	-4.76
Acme Precision Mfg	91.89	91.07	-0.82	90.03	-1.86
NovaCast Engineering	90.61	90.21	-0.40	88.04	-2.57
VoltEdge Components	87.60	87.51	-0.09	85.13	-2.47
TechForge Industries	84.74	84.65	-0.09	82.28	-2.46
Reliable Parts Co	83.32	82.68	-0.64	82.07	-1.25
AlphaForge Ltd	79.77	79.77	+0.00	74.41	-5.36
SteelPath Industries	79.52	78.14	-1.38	78.40	-1.12
Global Fabrication Ltd	77.02	76.06	-0.96	0.00	-77.02
PeakMetal Solutions	70.02	67.98	-2.04	70.24	+0.22

6.3 Performance Benchmarks

Metric	Target	Measured	Status
score_suppliers() — 10 suppliers	< 50 ms	~2 ms	✓ 25× faster
Unit test suite — 501 tests	All pass	501 / 0 fail	✓ 2.50 s
Expired cert detection	score=0	0.00 confirmed	✓
Zero cert detection	score=0	0.00 confirmed	✓

Metric	Target	Measured	Status
Injection stripping — 7 patterns	All covered	All + tested	✓ 24 unit tests
LLM schema validation (5 schemas)	Reject bad	Pydantic v2 enforced	✓
Evidence ID hallucination filter	Remove unknown IDs	filter_evidence_ids()	✓
Ranking calculation < 50 ms	< 50 ms	~2 ms	✓

7. Case Demonstrations

Case 1 — Successful: Clear Recommendation

Supplier: FastTrack Manufacturing (Mexico) | **Score:** 93.37 / 100 | **Confidence:** 87.6% — High

FastTrack wins the baseline on every cost and delivery dimension simultaneously. USMCA 0% duty + $4 truck freight produces the lowest landed cost ($112.00) among all compliant suppliers. 10-day lead time is fastest in the field. Triple certification (ISO 9001, IATF 16949, RoHS) leaves compliance at 100. Evidence retrieval completes in a single loop (coverage score 0.038 > 0.025 threshold). The AI recommendation cites 4 evidence chunks linking to exact PDF pages for price, lead time, and cert data.

Dimension	Score	Driver
Cost	95.80	Lowest landed cost ($112.00) among compliant suppliers
Quality	94.12	2.5% defect rate, 95% inspection pass, 4.5 customer rating
Delivery	86.15	10-day lead time (fastest), 93% on-time, 88% capacity
Risk	85.90	Financial 84/100, Supply 88/100 — near-shore resilience
Capability	100.00	All required capabilities + engineering support bonus
Compliance	100.00	ISO 9001:2015 + RoHS + IATF 16949 all valid
FINAL	93.37	Rank #1 of 10 suppliers

Case 2 — Ambiguous: Hidden Disqualifier in a Near-Tie

Conflict: FastTrack (93.37) vs Acme Precision Mfg (91.89) | **Gap:** 1.48 points | **Confidence:** ~72% — Medium

FastTrack wins mathematically on cost and delivery. Acme wins on quality (97.45 vs 94.12) and risk (94.40 vs 85.90). The gap is within the noise of any single data field varying. The system surfaces a hidden disqualifier: **Acme holds AS9100D aerospace certification; FastTrack does not**. If the end customer's quality plan requires AS9100D supplier approval, FastTrack is ineligible and Acme becomes the mandatory choice — regardless of score. The system explicitly flags this in the recommendation's *risks* and *assumptions* fields, reduces confidence to ~72% Medium, and shows a one-click sensitivity test: adding AS9100D as a required cert drops FastTrack to approximately #6 and makes Acme #1. This demonstrates the system escalating uncertainty to human judgment rather than hiding it.

Case 3 — Failure/Fallback: Expired Certification Detected

Supplier: PeakMetal Solutions (Vietnam) | **Final Score:** 70.02 (#10 of 10) | **Compliance Score:** 0

PeakMetal has the second-lowest landed cost ($108.10) — it would be a strong #3 contender. But its ISO 9001:2015 certificate expired 2024-12-31 (TUV SUD). The *is_valid=False* flag causes the mapper to exclude it from supplier_certs, making the compliance engine score 0. The system: (1) ranks PeakMetal last at 70.02 while continuing to rank all other 9 suppliers normally; (2) shows the specific cert name, issuer, and expiry date in the UI; (3) suggests the path forward — recertification + a cert-override scenario that shows PeakMetal would rise to ~#3 if recertified. Redis fallback is documented: if LLM times out, the deterministic ranking (PeakMetal last, reason visible) is always returned.

8. Intended Users, Assumptions, Limitations, and Human-Approval Points

8.1 Intended Users

User	Primary Need	System Output Used
Procurement Manager	Defensible recommendation + evidence	Dashboard · Scenario simulator · Executive report
Manufacturing Engineer	Cert verification · capability match	Supplier detail · Compliance scores · Evidence panel
Operations Manager	Supply chain resilience · lead time ranking	Risk breakdown · Scenario shipping sliders
Executive / Finance	One-page board summary	Generated PDF executive report (never logs in)

8.2 Seven Human-Approval Points

HAP	Trigger	Human Action Required
HAP-1	New supplier appears in ranking	Physical/3rd-party audit; legal verification; sanctions check
HAP-2	compliance_score = 0 (any supplier)	Confirm cert renewal status; upload updated certificate
HAP-3	Confidence < 60% (Low label)	Check which factor is low; upload missing documents
HAP-4	Score gap < 2 points between #1 and #2	Review head-to-head; run weight-sensitivity scenario
HAP-5	Scenario produces different #1 supplier	Assess likelihood of scenario conditions materialising
HAP-6	Before distributing executive report	Read in full; verify figures match internal understanding
HAP-7	Before issuing PO / letter of intent	Apply professional judgment + internal approval process

8.3 Key Limitations

• No live external data (no exchange rates, no sanctions checks, no credit APIs — all risk scores come from uploaded documents or the seeded dataset).

• Currency rates are static at evaluation time — not updated after document upload.

• Compliance check is binary (held / not held) — does not verify cert scope or manufacturing site.

• MOQ stored but not enforced in scoring — $2,000 MOQ for SteelPath is flagged in AI output but does not auto-disqualify.

• LLM confidence is narrative only — the 87.6% confidence figure is deterministic math, not LLM output.

• Self-RAG loop is bounded at 3 iterations — very sparse documents may still have thin evidence coverage.

• No multi-tenancy or authentication in demo configuration — single hardcoded project UUID.

9. Setup and Reproducibility

The full application can be reproduced locally in under 10 minutes with a free Gemini API key and a local PostgreSQL instance.

```
# Prerequisites: Python 3.12+, Node.js 18+, PostgreSQL 16+, Redis 7+

# 1. Backend
cd backend
python -m venv venv && venv\Scripts\activate
pip install -r requirements.txt
cp .env.example .env # set GEMINI_API_KEY + DATABASE_URL
alembic upgrade head

# 2. Seed database (10 suppliers, 3 scenarios, 17 document records)
cd ..
python scripts/seed_db_production.py
python scripts/generate_production_pdfs.py

# 3. Start backend + Celery worker
uvicorn app.main:app --reload --port 8000
celery -A app.workers.celery_app worker --loglevel=info

# 4. Frontend
cd frontend && npm install && npm run dev
# Open http://localhost:3000/dashboard

# 5. Run tests
cd backend
pytest tests/unit/ -v # 501 tests, 0 failures, 2.50 s
```

Environment Variables

Variable	Required	Description
GEMINI_API_KEY	Yes	Free from https://aistudio.google.com — used for LLM + optional embeddings
DATABASE_URL	Yes	PostgreSQL connection string (local or Supabase)
REDIS_URL	Yes	Redis connection (local or Upstash free tier)
OPENAI_API_KEY	No	Optional fallback LLM/embedding provider

10. Business Value and Competitive Differentiation

Human vs System — Time and Quality Comparison

Dimension	Manual Spreadsheet	Manufacturing Decision Copilot
Supplier evaluation time	2–4 hours	< 30 seconds
Landed cost calculation	Manual formula, error-prone	Deterministic engine, always correct
Expired cert detection	Missed until audit fails	Automatic — is_valid=False flag at ingestion
Risk assessment	Qualitative, subjective	5-factor weighted formula, per-supplier breakdown
Scenario analysis	New spreadsheet per scenario	Real-time re-ranking, < 2 ms

Dimension	Manual Spreadsheet	Manufacturing Decision Copilot
Recommendation audit trail	Email thread	Full evidence chain + audit_log in PostgreSQL
Executive summary	2 hours writing	AI-generated in < 3 seconds
Confidence quantification	None	Deterministic 5-factor score (0–100%)
Evidence traceability	None	Every AI claim → ChromaDB chunk → PDF page

Quantified Business Case

At 10,000 annual units, the FastTrack vs Acme landed cost difference ($9.75/unit) identified by the scenario engine represents **$97,500 in annual savings** — found in a 30-second scenario simulation that previously required a half-day spreadsheet exercise. The expired cert detection on PeakMetal (auto-scored 0) prevents a supplier qualification failure that typically costs $50,000–$200,000 in production delays and re-sourcing effort.

Why This Architecture Wins on Trust

Innovation	Deterministic engines + explainable AI — not a black-box chatbot
Traceability	Every recommendation links to exact PDF page + chunk ID in ChromaDB
Reproducibility	Same inputs → same scores, forever. Formula versioned (CALCULATION_VERSION='v1')
Resilience	AI failure → Redis cache → deterministic table. Never a blank screen.
Security	7 injection patterns stripped. Pydantic schema enforced on all LLM outputs.
Scalability	Pure Python engines scale linearly. ChromaDB + PostgreSQL scale horizontally.
Auditability	ai_requests · ai_responses · audit_logs tables capture every AI interaction.

11. Full Technology Stack

Layer	Technology	Version	Purpose
API Framework	FastAPI	0.141.1	Async REST API with Pydantic v2 validation
Runtime	Python	3.12+	Backend runtime
ORM	SQLAlchemy async + Alembic	2.0 / 1.14	PostgreSQL access + migrations
Database	PostgreSQL	16+	Source of truth for all structured data
Vector DB	ChromaDB	0.5.23	Semantic retrieval index (HNSW cosine)
Task Queue	Celery + Redis	5.4 / 5.2	Async document processing pipeline
AI Orchestration	PydanticAI	0.0.14+	Typed multi-provider LLM agents
LLM (default)	Gemini 2.0 Flash	google-genai 0.1	Free tier; fast inference
LLM (fallback)	OpenAI GPT-4o / Groq / Ollama	openai 1.67+	Provider failover chain
Embeddings	BAAI/bge-small-en-v1.5	local	384-dim, runs fully offline
Keyword Search	rank-bm25 BM25Okapi	0.2.2	Lexical retrieval for RRF fusion
PDF Parsing	PyMuPDF	1.24.14	Text + structure extraction
Frontend	Next.js 15 App Router	15+	React server components, App Router
UI Components	shadcn/ui + Tailwind CSS v4	latest	Accessible enterprise component library
State	TanStack Query v5 + Zustand v5	5+	Server state + global UI state
Charts	Recharts	v2	Score breakdown visualisations
Animations	Framer Motion	v11	DonutGauge, entrance animations
PDF Reports	@react-pdf/renderer	latest	Client-side executive report generation
Testing	pytest + pytest-asyncio	8.3 / 0.24	501 unit tests, 0 failures

12. Submission Deliverables Index

#	Required Deliverable	Document
1	Working end-to-end prototype	https://manufacturing-decision-copilot-farhan.vercel.app/dashboard
2	Source code + reproducible setup	submission/02_data_source_manifest.md §5
3	Architecture and data-flow explanation	submission/01_architecture_and_dataflow.md
4	Data/source manifest	submission/02_data_source_manifest.md
5	Baseline comparison and quantitative evaluation	submission/03_evaluation_results.md
6	Success · Ambiguous · Failure/Fallback cases	submission/04_case_demonstrations.md
7	Users · Assumptions · Limitations · HAPs	submission/05_users_assumptions_limitations.md
8	Presentation: decision, evidence, business value	submission/06_presentation_script.md
—	This PDF report (all sections combined)	submission/SUBMISSION_REPORT.pdf

Manufacturing Decision Copilot turns a 3-hour manual procurement evaluation into a 30-second evidence-backed recommendation that every stakeholder can read, challenge, and approve with confidence.